The Gravitational Field-Relation Operator

Registered Construction and Certification of Parity, Handoff, and Signature Geometry

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Revised methods formulation – First iterative pass – July 2026

Abstract

The Gravitational Field-Relation Operator (GFRO) is an equation-first method for converting a declared gravitational relation into an implicit geometric object and a certification ledger. The original formulation compared analytically assigned weak-field source and context contributions. The present revision replaces that scalar-first ontology with a registered evidence-object framework. A full scene and one or more intervention scenes are solved, their tensor fields are identified by declared point and frame maps, and relation operators are applied only after registration. For source ablation, the primary evidence object is the registered electric-Weyl response

$$\Delta_a E_{ij} = E_{ij}^{(\mathrm{full})} - \widetilde{E}_{ij}^{(-a)},$$

with Frobenius amplitude under a declared reference metric as the default scalarization. GFRO then forms continuous residuals, emits zero or threshold sets directly, and records root confidence, topology, registration, masks, and sensitivity in compact ledgers. The method is general enough to construct source-context GCS surfaces, source-source signature parity faces, handoff loci, degeneracy sheets, and fixed-skeleton relabeling boundaries. Analytic M/r^3 and source-summed weak-field tidal branches are retained as legacy correctness floors, not as the primary physical lane. The paper also elevates a key implementation lesson: tensor relation residuals should be built from continuous invariants or inner products whenever eigenbasis switching can create artificial discontinuities. GFRO does not decide which gravitational relation is physically meaningful; it supplies the disciplined machinery by which a declared lawful relation becomes an inspectable geometric object.

Keywords: general relativity; electric Weyl tensor; registered ablation; Frobenius norm; gravitational coherence surface; parity networks; implicit surfaces; tensor invariants; root certification; topology-ready ledgers.

1 Introduction

General relativity provides solved geometries, not generally source-owned parcels of curvature. A multi-source field may be seamless while finite interventions still produce structured, measurable differences. The revised Gravitational Coherence Surface program therefore treats source attribution operationally: solve the full scene, solve a comparison scene after a declared intervention, register the two geometries, and examine the resulting tensor response.

This creates a second problem. Once a lawful gravitational relation has been declared, how should its geometric level sets be constructed and certified?

Dense sampling offers one answer. A field can be evaluated over a large ledger, after which candidate surfaces, seams, and pockets are mined from the sampled values. That workflow is lawful but indirect.

It treats the whole sampled domain as a discovery substrate even when the object of interest is already specified by an equation.

GFRO adopts an equation-first grammar:

$$\boxed{\text{registered evidence objects} \rightarrow \text{continuous residual} \rightarrow \text{certified level set} \rightarrow \text{topology-ready ledger}}$$

The residual defines the object. A root or level-set solver emits candidate coordinates. The ledger verifies the declared relation at those coordinates and stores the information required to interpret, connect, and test the object.

This revision preserves the operator method while replacing its former source-field ontology. The original scalar and weak-field tidal branches remain valuable as analytic and historical validation lanes. The modern default begins with registered electric-Weyl ablation and uses the Frobenius norm as the primary direction-agnostic amplitude.

2 Scope and claim discipline

GFRO is not a new field equation, force law, causal horizon, material membrane, or modification of general relativity. It does not create physical structure by solving a residual. It constructs diagnostic geometry from a relation that must already be physically and mathematically declared.

The paper makes four methodological claims:

1. A declared gravitational relation can be represented by a continuous residual whose zero or threshold set defines a geometric object.
2. Direct level-set emission can replace dense volumetric mining as the primary discovery mechanism in suitable cases.
3. Compact ledgers remain essential for registration, residual substitution, topology, uncertainty, and claim status.
4. Tensor residuals require continuity discipline; eigenvectors may be useful interpretively while continuous invariants or inner products are safer computational primitives near degeneracy.

The revised framework reaches registered tensor ablation at the level of initial-data or slice comparisons. It does not yet claim complete dynamical, foliation-independent, or observer-independent extraction across arbitrary spacetimes.

3 Registered scenes and evidence objects

Definition 1 (Declared registered scene)**.** *A declared registered scene is the tuple*

$$\mathfrak{S} = (\Omega, \mathcal{R}, \mathcal{G}, \mathcal{I}, \gamma_{\mathrm{ref}}, \mathcal{O}, \mathcal{L}),$$

where Ω is the interrogation support, $\mathcal{R}$ is the source and intervention roster, $\mathcal{G}$ is the family of solved geometries or field bundles, $\mathcal{I}$ is the registration convention, γ_{ref} is the reference spatial inner product, $\mathcal{O}$ is the relation-operator family, and $\mathcal{L}$ is the certification ledger.

The declaration separates three layers that must not be collapsed.

Physical construction. The full and intervention scenes are generated under a stated approximation, initial-data family, matter model, coordinate chart, boundary condition, and mask convention.

Registration. Points and frames are identified between solved scenes before tensor subtraction or comparison. Registration may be physical, gauge-fixed, shared-coordinate, or otherwise declared, but it may not remain implicit.

Relation operator. A continuous scalar or tensor-derived residual is constructed only after the evidence objects inhabit a common comparison support.

Let Σ denote the full-scene slice and Σ_{-a} the slice obtained after ablating source a. A point identification is

$$I_a : U \subset \Sigma \to U_{-a} \subset \Sigma_{-a},$$

and a frame map is

$$P_a(x) : T_x\Sigma \to T_{I_a(x)}\Sigma_{-a}.$$

For a covariant spatial tensor $E^{(-a)}$, define the registered comparison tensor

$$\widetilde{E}^{(-a)}_{ij}(x) = P_i{}^k(x) P_j{}^\ell(x) E^{(-a)}_{k\ell}(I_a(x)) \,. \tag{1}$$

The registered ablation response is

$$\Delta_a E_{ij} = E^{(\text{full})}_{ij} - \widetilde{E}^{(-a)}_{ij}. \tag{2}$$

The response belongs to the comparison between solved scenes. It is not asserted to be a privately owned field component of source a.

4 Generalized GFRO definition

Definition 2 (Registered evidence object)**.** *A registered evidence object is any scalar, tensor, label field, complex, or derived channel whose comparison support, registration, reference metric, masks, and claim status have been declared.*

Let X and Y be registered evidence objects and let $\mathcal{O}_X$, $\mathcal{O}_Y$ be declared operators producing comparable scalar fields. Define the GFRO residual

$$\Psi_{\mathcal{O}}[X|Y](x) = \mathcal{O}_X[X](x) - \mathcal{O}_Y[Y](x). \tag{3}$$

The emitted parity object is

$$\Sigma_{\mathcal{O}}[X|Y] = \{x \in \Omega : \Psi_{\mathcal{O}}[X|Y](x) = 0\}. \tag{4}$$

Threshold objects are obtained by replacing zero with a declared tolerance or confidence condition.

For a spatial tensor A_{ij}, define the Frobenius amplitude under γ_{ref}:

$$\|A\|_{F,\text{ref}} = \left(\gamma^{ik}_{\text{ref}}\gamma^{j\ell}_{\text{ref}} A_{ij} A_{k\ell}\right)^{1/2}. \tag{5}$$

The default source-conditioned signature amplitude is

$$S_a(x) = \|\Delta_a E(x)\|_{F,\text{ref}}, \tag{6}$$

and the registered ablated-scene contextual amplitude is

$$C_a(x) = \|\widetilde{E}^{(-a)}(x)\|_{F,\text{ref}}. \tag{7}$$

The source-context GCS residual is therefore

$$\Psi_a^{\mathrm{GCS}}(x) = S_a(x) - C_a(x), \tag{8}$$

while many-source signature parity uses

$$\Psi_{ab}^{\mathrm{sig}}(x) = S_a(x) - S_b(x). \tag{9}$$

Equations (8) and (9) demonstrate that GFRO is a method for extracting declared relations, not a single physical operator.

5 Operator taxonomy

The revised framework organizes operators by evidence class rather than by a single scalar ladder.

Table 1: Modern GFRO operator classes.

Class	Representative residuals	Primary use
Amplitude relations	$S_a - C_a$, $S_a - S_b$, spectral-norm analogues	GCS parity, signature dominance, norm sensitivity
Tensor-shape relations	eigenvalue ratios, invariant polynomials, anisotropy measures	morphology and spectral-shape comparison
Orientation/ handoff	continuous tensor inner-product allegiance, registered frame-rotation channels	directional transition and handoff diagnostics
Topology/label relations	fixed-skeleton relabeling, born/removed strata, margin thresholds	relational-complex differences and retention
Scale/temporal relations	cross-resolution persistence, coarse-graining loss, future inter-slice residuals	mesoscale survival and dynamical extension

The tensor field remains the primary evidence object whenever available. Scalarization extracts one level-set family and must not be mistaken for the whole tensor response.

6 Regularity, continuity, and level-set meaning

Proposition 1 (Local regularity)**.** *Let $\Psi : \Omega \to \mathbb{R}$ be continuously differentiable. If $\Psi(x_*) = 0$ and $\nabla\Psi(x_*) \neq 0$, then the zero set is locally a smooth embedded codimension-one surface near x_*.*

Proof. The result follows from the implicit function theorem. The nonzero gradient guarantees that at least one coordinate derivative is nonzero at x_*, allowing the zero set to be represented locally as the graph of a differentiable function. □

Where $\nabla\Psi = 0$, the object may pinch, self-contact, branch, merge, or collapse into a lower-dimensional stratum. GFRO must record those failures of regularity rather than smoothing them away.

The same discipline applies to seams and nodes. For residuals $\Psi_1, \ldots, \Psi_m$ in three dimensions, a generic intersection has dimension $3-m$ only when the gradients are independent. Symmetry, degeneracy, masks, and boundary clipping can change this count.

7 Analytic correctness floor: the legacy R0 branch

The original scalar tidal-readable benchmark remains a useful implementation check. For source i,

$$Q_i(x) = \frac{M_i}{r_i(x)^3}. \tag{10}$$

Pairwise equality gives

$$\frac{M_i}{r_i^3} = \frac{M_j}{r_j^3}, \qquad \frac{r_i}{r_j} = \left(\frac{M_i}{M_j}\right)^{1/3}. \tag{11}$$

Equal masses produce a perpendicular bisector plane. Unequal masses produce an Apollonius sphere. This branch remains valuable because it offers analytic geometry against which root emission, residual substitution, coordinate conventions, and connectivity can be tested.

Its status is limited. Equation (11) is a proxy correctness floor, not the primary ontology of a fused nonlinear scene.

8 Legacy weak-field tensor benchmark

The former R2 branch assigned each point source a weak-field tidal tensor

$$E_i(x) = \frac{M_i}{r_i^3}\left(3\hat{r}_i\hat{r}_i^{\mathsf{T}} - I\right), \tag{12}$$

for which

$$\|E_i\|_F = \sqrt{6}\,\frac{M_i}{r_i^3}. \tag{13}$$

Pairwise FN parity therefore reproduces the R0 topology for isolated point-source contributions. Source-context comparison differs because the context tensor is summed before norming:

$$\Psi_{i|K}^{\text{legacy}} = \|E_i\|_F - \left\|\sum_{j\in K} E_j\right\|_F. \tag{14}$$

This lane remains useful for testing tensor cancellation, orientation sensitivity, and norm-after-summation effects. It does not equal registered ablation in nonlinear geometry and is therefore retained as a historical tensor benchmark only.

Table 2: Status of principal GFRO lanes.

Lane	Evidence object	Status
R0 analytic	M_i/r_i^3	Analytic proxy correctness floor
Legacy R2	Per-source weak-field tidal tensors	Historical tensor-summation benchmark
Primary modern	Registered $\Delta_a E_{ij}$	Default source-conditioned evidence object
Sensitivity	spectral norm, invariant polynomials, spectral shape	Robustness and branch comparison

9 Direct emission workflow

A modern GFRO extraction follows a fixed order.

1. Declare the full scene, intervention, domain, masks, physical regime, and solve method.
2. Solve the full scene and the comparison scene.
3. Register points and frames by I_a and P_a.
4. Form the tensor response $\Delta_a E$.
5. Apply the declared operator, using FN amplitude as the default scalar lane.
6. Construct the residual Ψ.
7. Emit candidate zero or threshold coordinates through a declared solver.
8. Re-substitute every emitted coordinate into the residual.
9. Estimate gradients, branch confidence, masks, and topology membership.
10. Repeat sensitivity lanes and promote only claims that survive the declared tests.

This workflow changes the rank of the ledger. The ledger does not need to discover the object by exhaustive sampling, although dense sampling may remain valuable for coverage tests. The equation defines the candidate object; the ledger certifies and contextualizes it.

10 Certification ledgers

A modern GFRO ledger should contain enough information to reproduce both the relation and its extraction.

Table 3: Recommended registered GFRO ledger schema.

Field class	**Purpose**
Full-scene identifier	Links every root to the solved parent scene.
Comparison-scene identifier	Identifies the ablated or otherwise intervened solve.
Intervention identifier	Names the source, group, roster change, or parameter change.
Point and frame maps	Records I_a, P_a, and any gauge-fixed correspondence.
Reference metric	Declares the inner product used for contraction and scalarization.
Tensor witness	Identifies electric Weyl, spatial Ricci in a declared slice, or another field object.
Operator/scalarization	Records FN, spectral, invariant, projection, margin, or label rule.
Coordinate and residual	Stores emitted location and residual value after re-substitution.
Gradient/root confidence	Supports local regularity, orientation, and root quality.
Spectral-gap confidence	Flags eigenframe instability near repeated eigenvalues.
Mask/support status	Records puncture, matter, boundary, active-support, and exclusion states.
Topology membership	Links coordinates to surfaces, seams, nodes, pockets, or complexes.
Sensitivity status	Records resolution, norm, registration, and perturbation outcomes.
Claim level	Separates definition, implementation, observation, benchmark, and hypothesis.

A rendered mesh is downstream of this record. Visual closure is not equivalent to numerical or topological closure.

11 Tensor-safe residuals and the eigenframe continuity hazard

The original GFRO validation program exposed an important failure mode. Direct principal-eigenvector residuals may jump discontinuously when eigenvalues cross or nearly coincide. A bracketed root solver can mistake such a basis switch for a true zero crossing.

Methodological principle 1 (Continuity rule)**.** *GFRO residuals should be formulated through continuous tensor invariants or tensor inner products whenever eigenbasis switching can create artificial discontinuities. Eigenvectors may remain useful interpretive objects, but they should not automatically define the computational residual.*

A tensor-safe handoff residual may compare allegiance through Frobenius inner products. For evidence tensors A, B, and a declared reference tensor T,

$$H[A|B;T] = \frac{(A:T)^2}{\|A\|_F^2} - \frac{(B:T)^2}{\|B\|_F^2}, \qquad A:T = A_{ij}T^{ij}. \tag{15}$$

For registered ablation signatures, one may substitute $A = \Delta_a E$ and $B = \Delta_b E$, with the reference tensor chosen and justified by the experiment.

Equation (15) does not produce a universal physical handoff surface. It demonstrates the implementation rule: preserve the intended directional comparison through continuous tensor operations whenever possible.

12 Topology, dominance, and retention

GFRO emits candidate level sets. Not every emitted set deserves promotion into a retained network or certified relational complex.

For many-source FN amplitudes S_i, define the closed dominance cell

$$\mathcal{C}_i = \{x : S_i(x) \geq S_k(x) \text{ for all } k\}, \tag{16}$$

and the strict interior

$$\mathring{\mathcal{C}}_i = \{x : S_i(x) > S_k(x) \text{ for all } k \neq i\}. \tag{17}$$

Candidate equality sets are

$$\Sigma_{ij} = \{x : S_i = S_j\}. \tag{18}$$

A retained ordinary face interior is the subset of Σ_{ij} on active support where i and j jointly dominate all remaining sources and no higher-order tie is present. Triple and higher-order ties are stored as separate lower-dimensional strata.

A retention functional may be written schematically as

$$\mathcal{W} = extsf{Retain}(\{\Sigma\}, \mathcal{L}, \mathcal{V}), \tag{19}$$

where $\mathcal{L}$ is the annotation ledger and $\mathcal{V}$ is the validation record. Retention may require active support, residual cleanliness, continuity, closure status, branch persistence, registration sensitivity, norm sensitivity, perturbation stability, and topology confidence. No universal retention functional is claimed.

13 Completeness and negative certification

Direct emission changes the discovery workflow but does not eliminate completeness questions.

Positive certification establishes that emitted coordinates satisfy the relation. It includes residual substitution, gradient checks, branch identification, connectivity, and registration integrity.

Coverage testing asks whether relevant branches were missed. Multi-axis emission, adaptive seeding, ray sweeps, box sweeps, seam-specific extraction, and reconciliation with dense audits can reduce blind spots.

Negative completeness asks whether no unreported roots exist in a declared region. This requires stronger methods such as interval arithmetic, certified residual bounds, box exclusion, or other validated numerics. A visually empty region is not proof of absence.

Single-axis solvers can miss surfaces tangent to the solve direction. Surface extraction can under-resolve seams and nodes. Mask boundaries can manufacture apparent openings. These are engineering and mathematical gaps to be reported, not hidden.

14 Historical S4 validation object

The original GFRO paper used a four-source anisotropic weak-field scene and a weak source, S4, as a stress-tested validation object. Under the legacy R0 and source-summed R2 lanes, the S4 pocket was reported as closed, residual-clean, perturbation-stable, tensor-deformed, and annotated by entropy, cancellation, and orientation diagnostics.

Those results retain methodological value. They demonstrate equation-first emission, raywise closure auditing, residual substitution, perturbation testing, tensor cancellation analysis, and the discovery of eigenbasis discontinuity hazards. They do not validate registered ablation or curvature-native GFRO.

The historical numerical values therefore belong only to the declared proxy and source-summed weak-field scene. In the revised program, S4 should be read as a legacy implementation benchmark whose strongest surviving contribution is the validation discipline it forced.

The binary attribution entropy result is similarly retained as a consistency relation. If

$$w_A = \frac{R_A}{R_A + R_K}, \qquad w_K = \frac{R_K}{R_A + R_K},$$

then

$$S_{\mathrm{attr}} = -w_A \ln w_A - w_K \ln w_K$$

is maximized at $w_A = w_K = 1/2$, exactly the parity condition. This is an information-theoretic restatement of binary equality, not an independent physical witness.

15 Evidence and claim register

Table 4: Current claim status after revision.

Claim	Status	Scope
Residual level sets define diagnostic geometry	Defined	Mathematical construction under declared regularity
R0 analytic recovery	Demonstrated	Proxy correctness floor
Equation-first emission and residual ledgers	Demonstrated	Legacy weak-field implementations
Registered ablation GFRO formulation	Defined	Modern default framework
FN source-context and source-source operators	Implemented in revised GCS program	Requires per-experiment registration declaration
Tensor-safe inner-product handoff rule	Corrected benchmark	Continuity safeguard, not universal physical handoff
Curvature-native GFRO extraction	Pending direct certification	Static-slice and broader GR lanes remain to be run
Universal retention functional	Open	Scene and purpose dependent
Negative completeness certification	Open	Requires validated interval or box methods
Temporal GFRO	Future	Requires registered slice sequence and causal discipline

16 Limits and research agenda

The revised paper remains limited in several ways. Registration is declared rather than unique. Electric-Weyl comparisons depend on slicing and observer choice. Static initial-data demonstrations do not establish dynamical persistence. FN is a useful default but not a complete tensor description. The historical S4 result belongs to a weak-field source-decomposition lane. Negative completeness remains open. No observational detection is claimed.

The immediate research agenda is:

1. implement curvature-native registered GFRO on exact static fused scenes;
2. compare FN, spectral, and invariant-polynomial residuals;
3. certify multi-axis surface coverage and lower-dimensional seam extraction;
4. integrate fixed-skeleton relational-complex differences;
5. test retention across resolution, registration, norm, and perturbation lanes;
6. develop interval or box methods for negative completeness;
7. extend the grammar to registered temporal slice sequences without conflating slice comparison with causal evolution.

17 Conclusion

GFRO survives the evolution of the GCS program because its durable contribution is methodological, not tied to any one proxy field. The operator converts a declared gravitational relation into a continuous residual, an implicit geometric object, and a certification ledger.

The revised ontology begins with solved scenes and registered evidence objects. For source ablation, the primary object is the tensor response $\Delta_a E_{ij}$; the Frobenius norm supplies the default direction-agnostic amplitude; and GFRO extracts source-context, source-source, handoff, topology, or scale relations according to the declared operator.

Analytic M/r^3 geometry and source-summed tidal tensors remain valuable as correctness floors and historical benchmarks. They no longer define the physical center of the method. Likewise, S4 remains useful as a validation lesson rather than as evidence for the modern registered-ablation lane.

GFRO does not decide which gravitational relation is physically meaningful. It provides the disciplined machinery by which a declared lawful relation becomes an inspectable geometric object.

A Compact R0 derivation

For $Q_i = M_i/r_i^3$ and $Q_j = M_j/r_j^3$, parity implies

$$M_i r_j^3 = M_j r_i^3,$$

therefore

$$\frac{r_i}{r_j} = \left(\frac{M_i}{M_j}\right)^{1/3}.$$

Equal masses yield the perpendicular bisector plane. Unequal masses yield an Apollonius sphere.

B Minimal extraction pseudocode

```
input: full scene, intervention scene, registration, operator, domain
solve full and comparison scenes
register comparison tensor onto full-scene support
construct evidence objects
construct continuous residual Psi
emit candidate roots / level-set coordinates
for every emitted coordinate:
    resubstitute Psi
    estimate gradient and confidence
    record masks, branch, topology, and claim status
repeat declared sensitivity lanes
retain only certified objects
```

C Non-claims

This paper does not claim modified gravity, a new force, a physical wall, a causal or trapping horizon, source ownership of a curvature parcel, a universal scalarization, a universal registration, complete root coverage, observational detection, or a finished wireframe retention law.

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